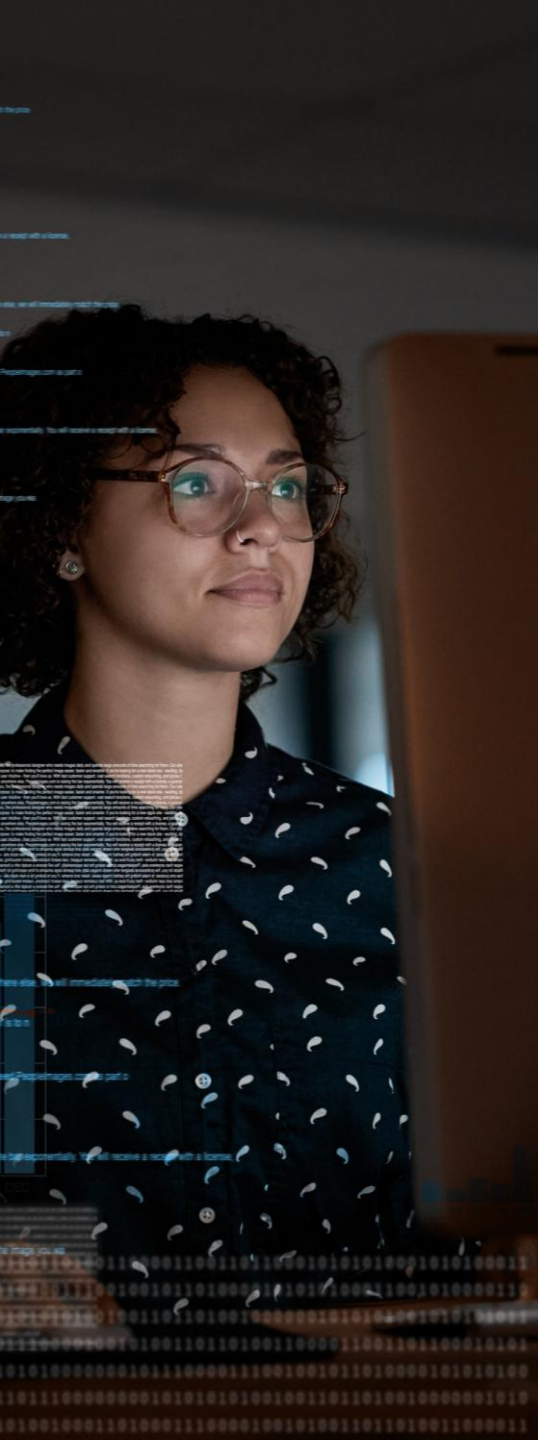




PMI-ACP® EXAM PREP

PMI Authorized Training Partner

BOOTCAMP

Session 7

- This course will assist learners in preparing for PMI's PMI-ACP Exam (2024 Update)

ATTENDANCE TRACKING

Percipio Users:

Name is based on your log in information in Percipio

Using Zoom:

Enter your first and last name

BREAKS



Yes! We will have periodic breaks

For attendance purposes, please stay logged in during all breaks.



CHAT vs Q&A

Please use the **Chat** for:

- **Greetings** before the session starts and during breaks
- Once the session starts , the chat may be closed or changed to *Hosts & Panelists Only* to minimize disruptions and focus on important information.
- The instructor may open the chat during the session for student **to respond to the instructor's questions** and create a group dialog.

CHAT vs Q&A

Please use the **Q&A** for:

- **Technical assistance** – Begin with: Percipio or Non-Percipio student
- Guidance on how to **access course material** – Begin with: Percipio or Non-Percipio
- Clarification and **questions on lecture points**, if not answered by instructor
- The Q&A may be open and closed throughout the session to allow us to address questions/issues in a timely manner.
- **Please be very patient, the support team responds to many inquiries per session**

IS LIVE ATTENDANCE REQUIRED?

- **YES**, if you are taking this training to register for the PMI-ACP exam
- You are **allowed to miss up to two sessions IF** you make up the sessions by **watching the video replays**.
- A **missed** session means you were disconnected for **more than a total of 15 mins** for the duration of the session.
- *Please see the Bootcamp Calendar for information about upcoming sessions at: <http://calendar.skillsoft.com/>



ACCESSING THE VIDEO REPLAYS

1. Go to: <https://github.com/Skillsoft-Content/PMI-ACP-Replay>
2. Replays will be available within 2 business days after the session ends.
3. Click on the Excel file for the year you attended the Bootcamp. You won't see a *file open* option, but it is selected.
4. Click the *Download raw file* button on the far left-hand side.
5. Open the downloaded file using this password: **acpB00tcampReplay!**



Those are zero's not the letter O. The password is case sensitive.

6. Locate and open the worksheet tab that corresponds with the bootcamp you attended
7. Make a note of the passcode.
8. Paste the provided link into your browser.
9. Complete the required registration steps
10. Input the passcode when prompted
The password to open the Excel file is NOT the passcode to access the replay.

Note: Replays will be available for 1 year.
They are not available for download.

No limit to watch replays to study

PMI®-Authorized PMI Agile
Certified Practitioner (PMI-ACP)®
Exam Preparation Course

Lesson Four

Delivery

Version 1.0 | 2024 Release



During this segment



**Seek early
feedback**



**Manage
agile metrics**



**Manage
impediments
and risk**



**Recognize
and eliminate
waste**



**Perform
continuous
improvements**



**Engage
customers**



**Optimize
flow**



**Determine which
metrics are
appropriate for a
given audience**



**Radiate metrics
across the relevant
audience**



**Review and
analyze metrics**



**Use metrics
insights for
decision-making**



**Seek early
feedback**



**Manage
agile
metrics**



**Manage
impediments
and risk**



**Recognize
and eliminate
waste**



**Perform
continuous
improvements**



**Engage
customers**



**Optimize
flow**



Use metrics insights for decision-making

Section 4 of 4

Velocity

A measure of the average amount of work a team can complete in one iteration



Velocity trends



**Project
schedule**



Potential issues



**Realistic
goals**

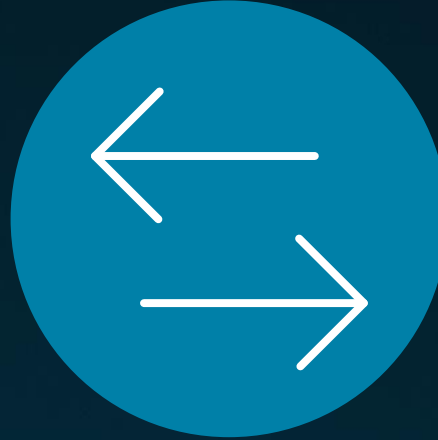


**Improve team
performance**

Challenges with velocity



**Easy to
manipulate**



**Not easily
compared**



Internal use

Use CSAT scores to improve customer satisfaction

Identify areas where customers are not satisfied with the product or service being delivered



Use CSAT scores to evaluate team performance

Evaluate the team's performance in delivering a product or service that meets customer needs

Defect density

The number of defects found during a period of development divided by the size of the overall project



Using defect density to help make decisions



**Identifying
areas for
improvement**



**Comparing
projects**



**Quality
goals**



**Evaluating team
performance**

Eliminating bottlenecks

When people performing a specific step in the development process can't keep up with the amount of incoming work



Spotting bottlenecks

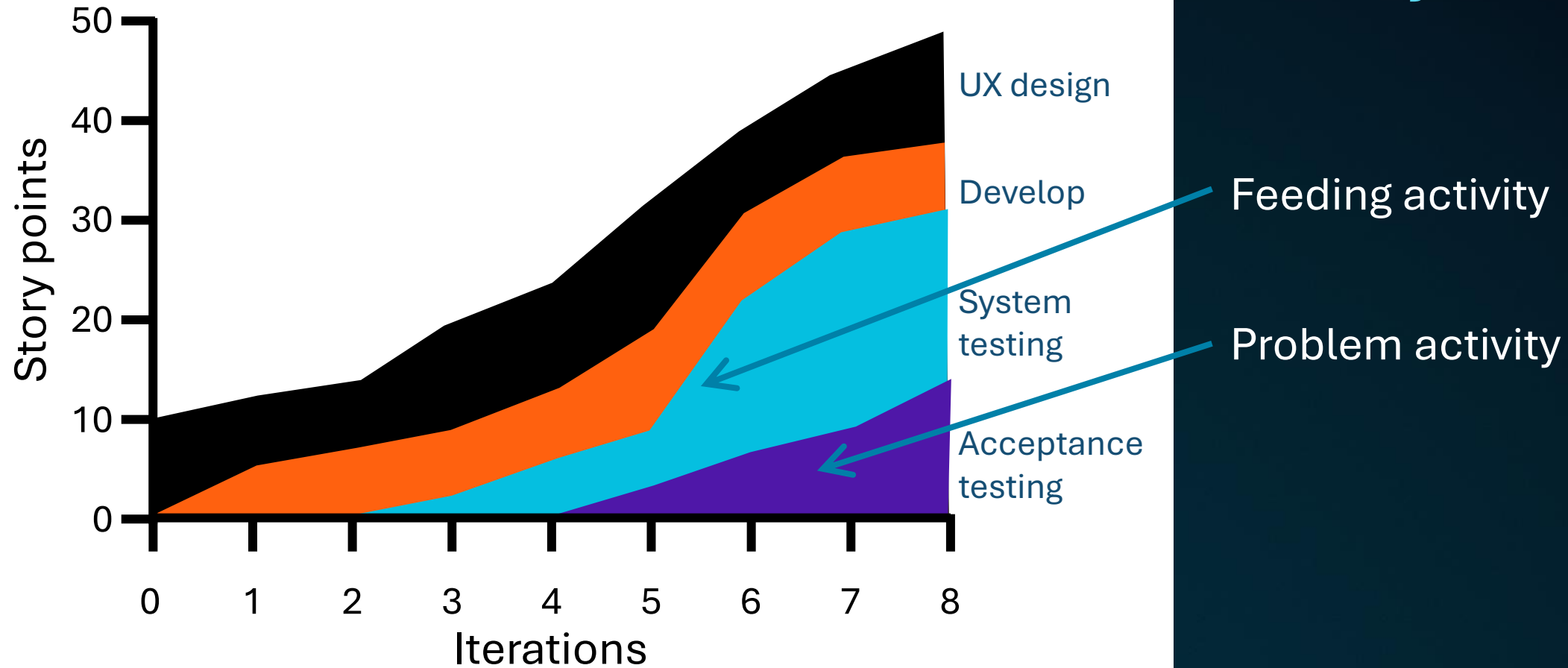


**Cumulative flow
diagrams**



**Value stream
mapping**

Using a cumulative flow diagram to identify a bottleneck



What is a value stream?

A **value stream** is an organizational construct that focuses on the flow of value to customers through the delivery of specific products or services.

Value stream: Buying a cup of coffee



~~The customer enjoys the drink~~

The customer enjoys the drink

Steps in making a value stream map



**Identify the
actions**



Time worked on



Time waiting



Time between steps



Loopbacks



Visual

You can do the same thing at your organization

Value stream maps make the work taking place visible

This makes it possible for people to have intelligent conversations

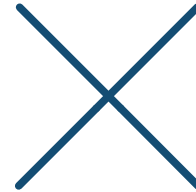


Knowledge check

True or false: The value stream begins and ends with the organization.



True



False

Knowledge check

What might the inflection at point A in the cumulative flow diagram shown below indicate?



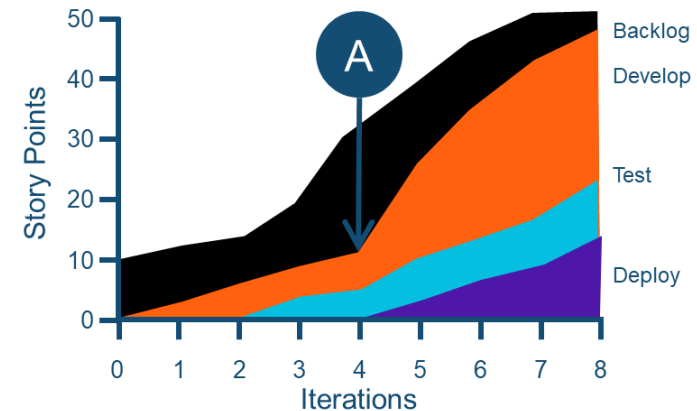
Excess capacity in test



A bottleneck in test



A bottleneck in deploy



The sharp increase in the amount of work in **Develop** could indicate a **bottleneck in Test**. It appears Test is falling behind since it is the activity below the widening area.



**Proactively
identify risks
and
impediments**



**Engage the
team to find the
most
appropriate
course of
action**



**Prioritize
impediment
removal and
risk mitigation
activities**



**Monitor and
control risks
and
impediments**



**Use lessons
learned to
avoid risk and
impediment
recurrence**



**Seek early
feedback**



**Manage
agile metrics**



**Manage
impediments
and risks**



**Recognize
and eliminate
waste**



**Perform
continuous
improvement**



**Engage
customers**



**Optimize
flow**



Proactively identify risks and impediments

Section 1 of 5



What is risk?

Risk is an uncertain event or condition that—if it occurs—will have a positive or negative effect on one or more of the project's objectives.

Risks can be positive and negative



Risks with positive outcomes
are called opportunities



Risks with negative outcomes
are called threats

What is risk management?

Risk management is the processes of identifying and analyzing project risks and determining appropriate responses.

Agile is not a risk management approach

Building something iteratively
does not ensure
risks are managed



Risk
management
approach



Layering risk management into agile

Agile approaches offer many cycles and mechanisms that can be used for risk management

Identifying risks and impediments



**Collaborative
discussions**



Expert judgment



**Formal
risk session**



**Informal
risk session**



Interviews



**Patterns and
common lists**



**SWOT
analysis**

Collaborative discussions

Wide range of opinions

Discussion needs to be facilitated

May not be willing to publicly discuss some risks



Expert judgment

Quick

May not have access

Inexperienced teams may
ignore risks identified by
experts

Formal risk session

Makes risks visible

Useful in disagreement

Regulatory compliance

Advance preparation

Difficult to schedule

Rubber-stamping





Informal risk session

Inclusive

Minimal preparation

Difficult to schedule

Applicable to regulatory
situations

Interviews

Private discussions about risks that people may not be willing to discuss openly

Miss risks when not discussed as a group





Patterns/ common lists

Reusing lists

New types of risks may
be missed

SWOT analysis

Identifies positive risks

Competitive situations

Collaborative
group discussions

Goes beyond threat
identification

Can take time





Engage the team to find the most appropriate course of action

Section 2 of 5

Why engage the team?

Better decisions

Promotes problem-solving

No endless appeals

Fosters action



Source: Wondolleck, J., & Yaffee, S. (2000). *Making collaboration work: Lessons from innovation in natural resource management*. Island Press.



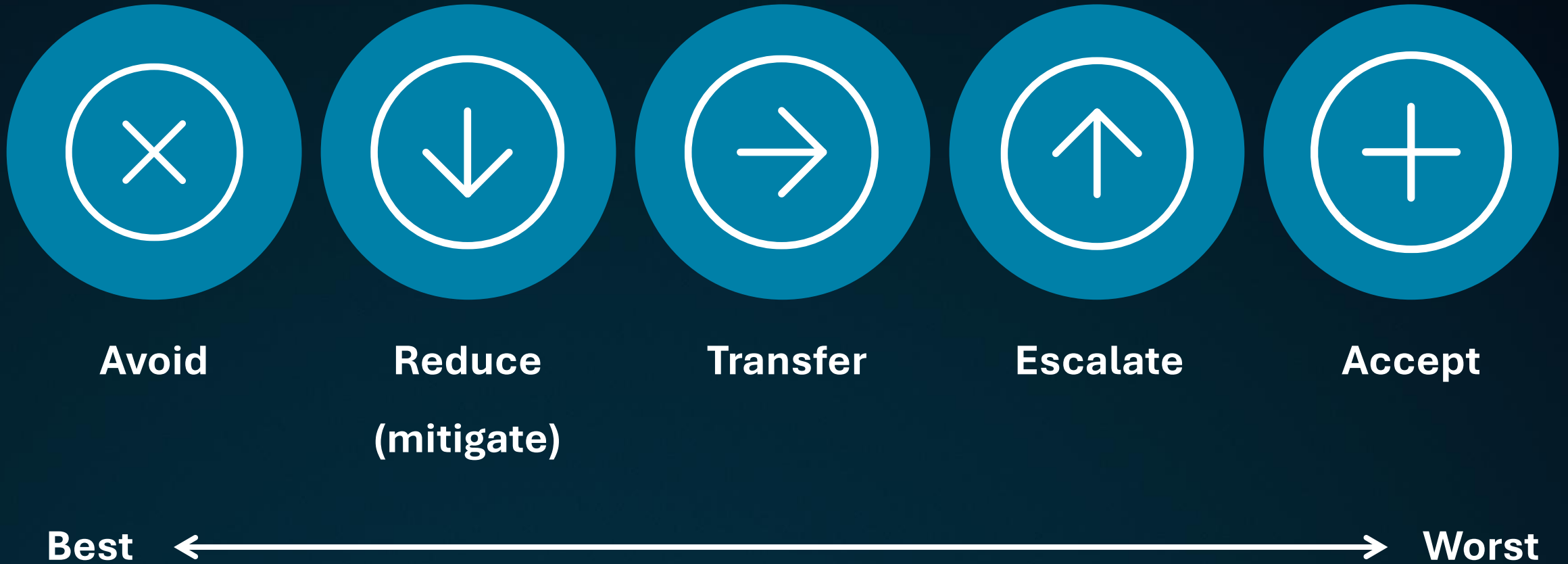
© 2024 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI course.

Collaboration isn't magic

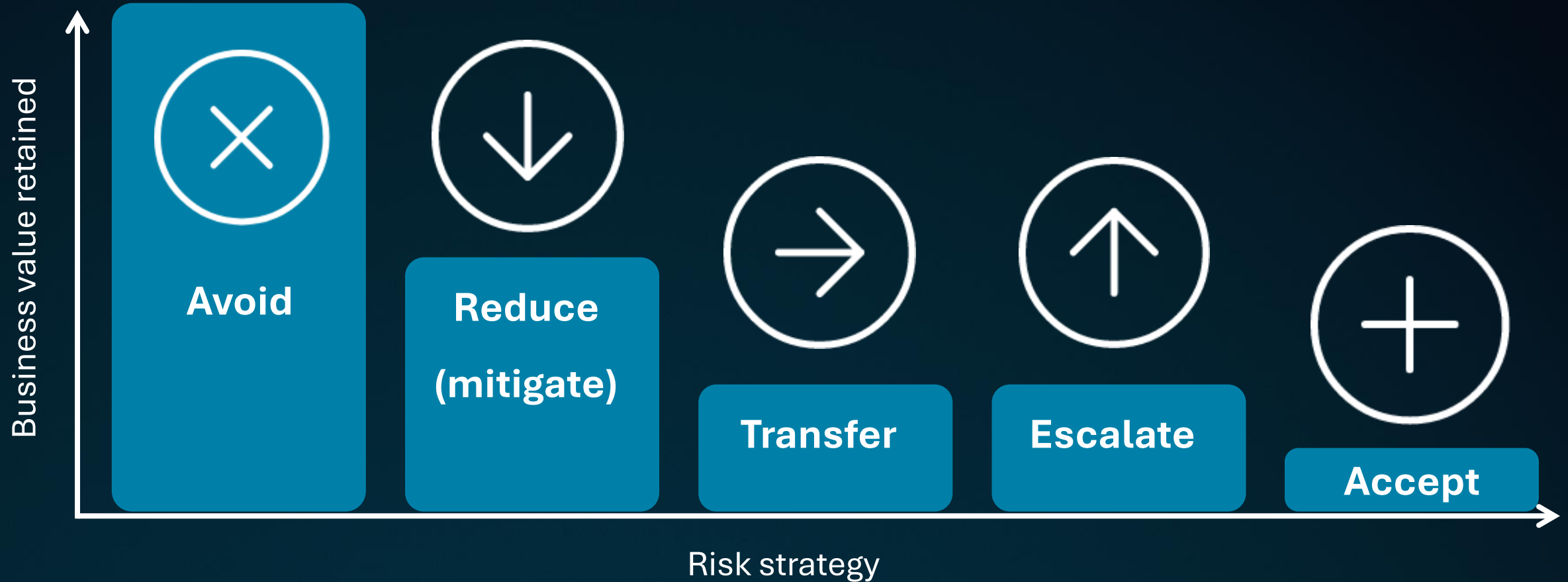
There are downsides and
potential for misuse

The benefits far outweigh
any potential liabilities

Planning risk responses



Business value retained



Denial

Not a valid response

Can be catastrophic



Spike

A **spike** is a short time interval within a project, usually of fixed length, during which a team conducts research or prototypes an aspect of a solution to prove its viability.

Spikes to explore and reduce risk

Spikes typically take less time than a standard iteration





Short iterations

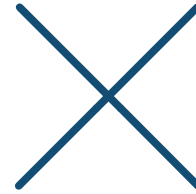
Allow us to check
frequently and ensure our
risk responses are working

Knowledge check

True or false: Denial is a valid response to risk



True



False

Knowledge check

True or false: Avoiding risk is preferable to accepting it



True



False



Prioritize impediment removal and risk mitigation activities

Section 3 of 5

Risk-adjusted backlog

Risk response activities

Collaborative discussion

Reduces amount of risk



Put risk responses into the team's backlog

Make sure risk responses
are a regular part of the
team's day-to-day work





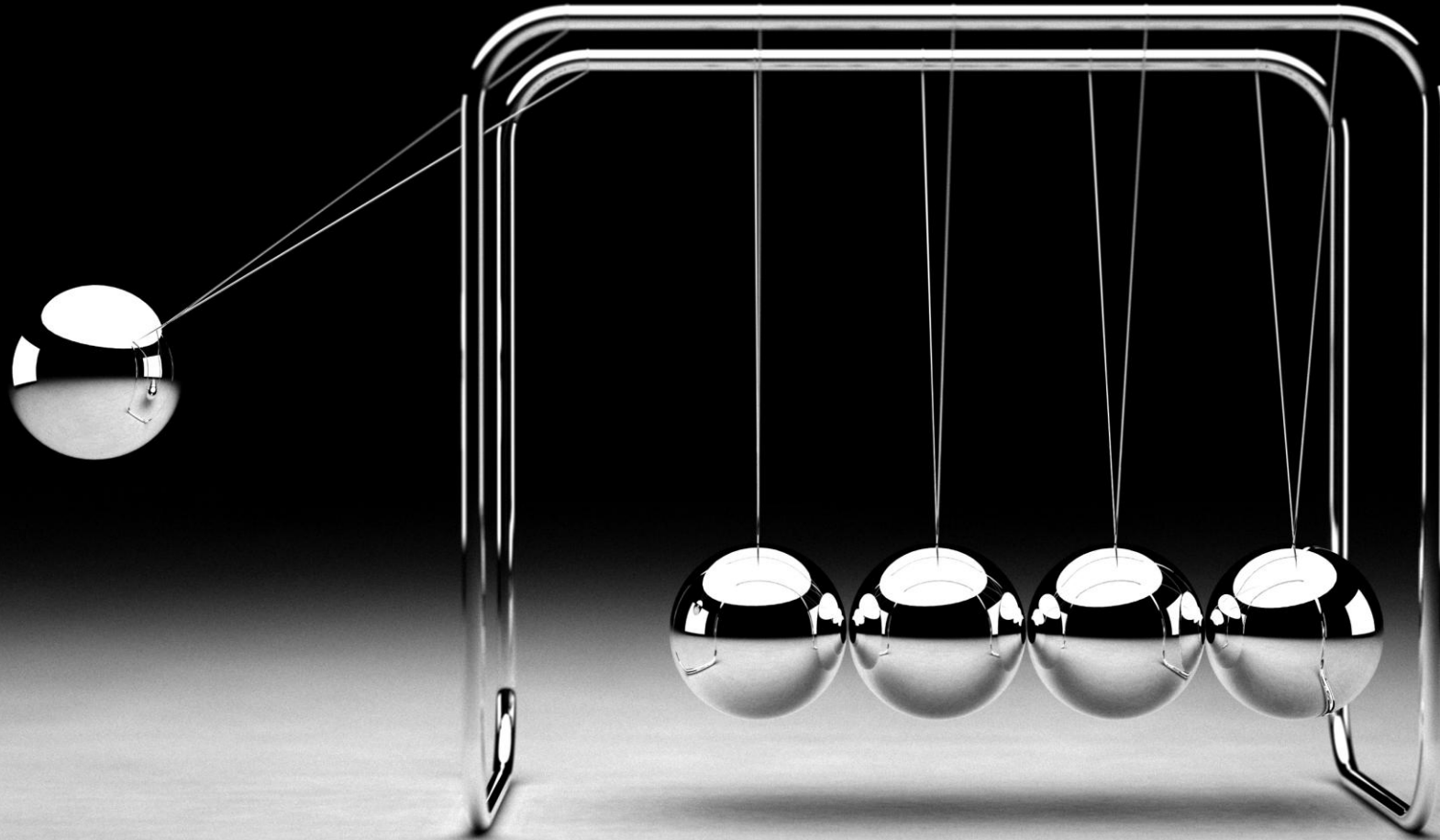
Residual risk

Risk that remains after your risk responses have been implemented

Should be clearly identified, analyzed, documented, and communicated

Secondary risks

How do we account for the effect secondary risks have on the primary risks that triggered them?



Adjusting the value of a risk response action

$$\text{Net EMV} = \text{EMV of residual risk} + \text{EMV of secondary risk}$$



Risk with EMV of US\$60,000

Response only addresses half the problem

Secondary risk with EMV of US\$10,000

$$\text{US\$60,000} \times 50\% = \text{US\$30,000}$$

$$\text{US\$30,000} + \text{US\$10,000} = \text{US\$40,000}$$

Add to the risk-adjusted backlog

Feature	US\$50,000
Feature	US\$40,000
Feature	US\$30,000
Feature	US\$20,000
Risk response	US\$20,000
Feature	US\$5,000

**Repeat the process for
each of the remaining
risk responses**



Work with the product owner to create the risk-adjusted backlog

Product owners rank the work based on its business value

That includes work undertaken to avoid, reduce, escalate, or transfer risk

Knowledge check

How do we prioritize impediment removal and risk mitigation activities ?

A. Discuss them in daily coordination meetings

B. Reach out to the product owner

C. Insert them into the backlog

D. Poll relevant stakeholders

The correct answer is **insert them into the backlog**. By doing so, we make sure our risk responses are not supplemental activities but a regular part of the team's day-to-day work.

Knowledge check

How do we account for the effect secondary risks have on the primary risks that triggered them?

A. Add the response to the risk-adjusted backlog

B. Estimate the net expected monetary value

C. Work with the product owner

D. Identify, analyze, document, and communicate residual risk

The correct answer is **estimate the net expected monetary value.**



Monitor/control risks and impediments

Section 4 of 5

Risk retrospectives

A team review of the project risks identified and risk management processes used



How does the team use this meeting?



Still active?



**Changes to
probability, impact,
or urgency**



**Effectiveness of
risk responses**



**Identify new risks;
close those no
longer active**

Risk retrospective questions



Are we eliminating or reducing our risks?

Do we have any new or increasing risks?

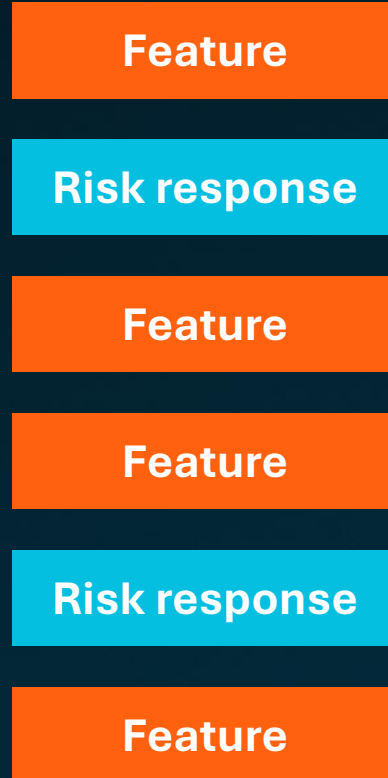
What are the root causes of our risks, and can we eliminate any of them?

Which risk strategies are working, and which are not?

For risks that we chose to transfer or escalate,
how are the third parties managing them?

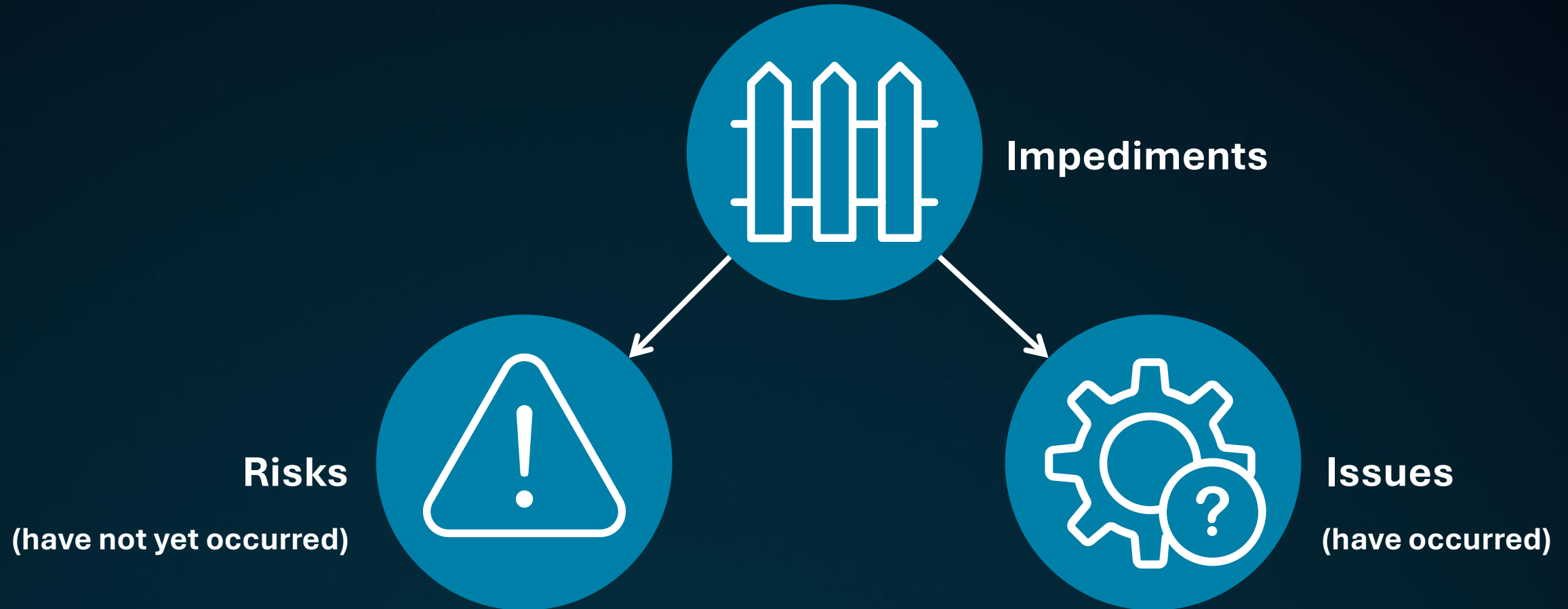
How are our team risk management capabilities developing?

Reprioritize the backlog



**The product owner should
use the results of your risk
retrospective to
reprioritize the backlog**

Impediments raised at daily coordination meetings



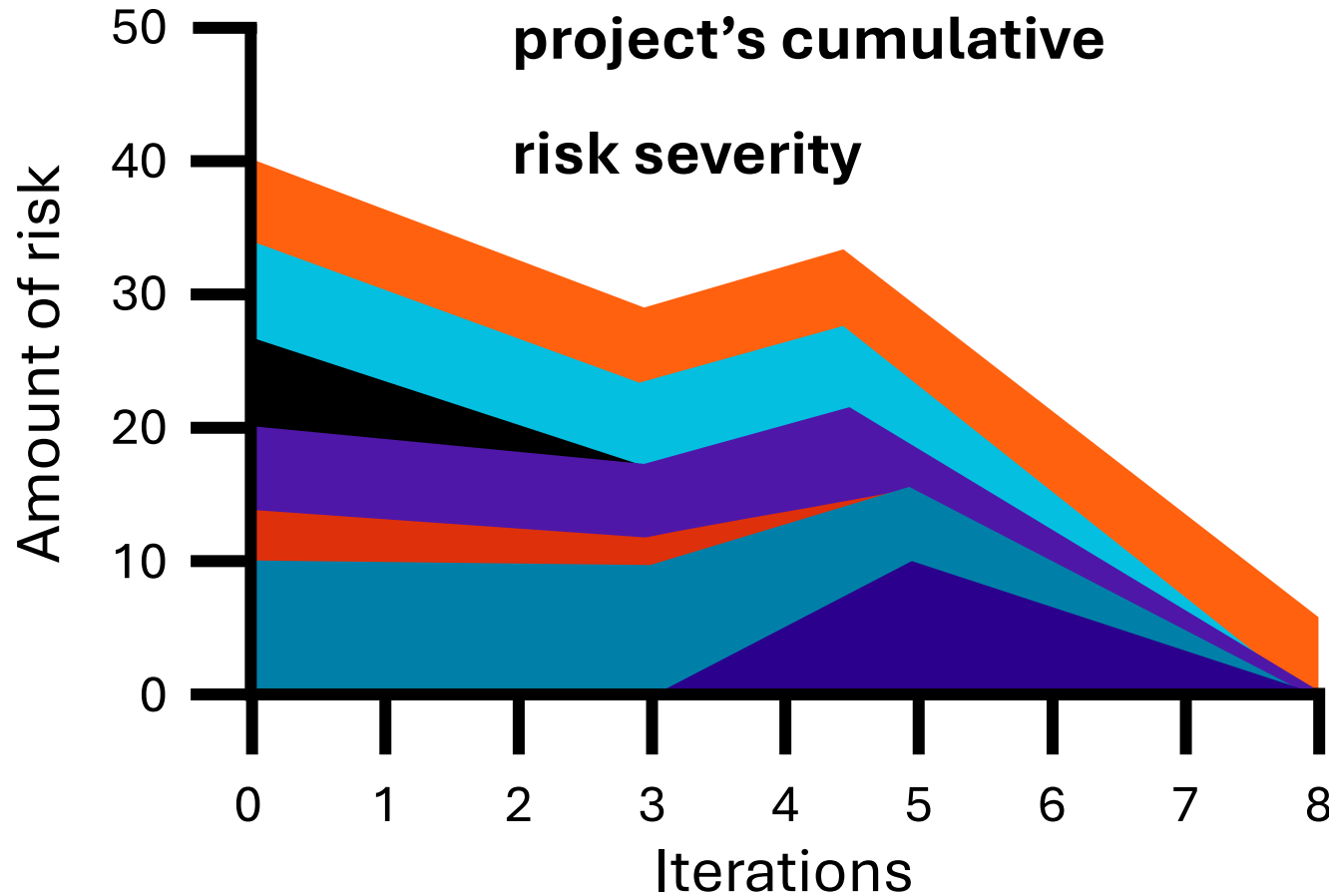


Visualizing risks

One tool that can help
make sense of all this data
is a risk burndown graph

What is a risk burndown graph?

A chart that shows the project's cumulative risk severity



- Risk 1
- Risk 2
- Risk 3
- Risk 4
- Risk 5
- Risk 6
- Risk 7

Risk management should be a team activity

We want risk management activities to be part of the day-to-day activities of a larger pool of project stakeholders



Knowledge check

True or false: Issues are risks that have already happened



True



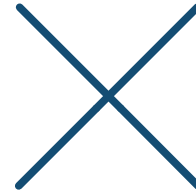
False

Knowledge check

True or false: Risk retrospectives should be held at the end of the project



True



False



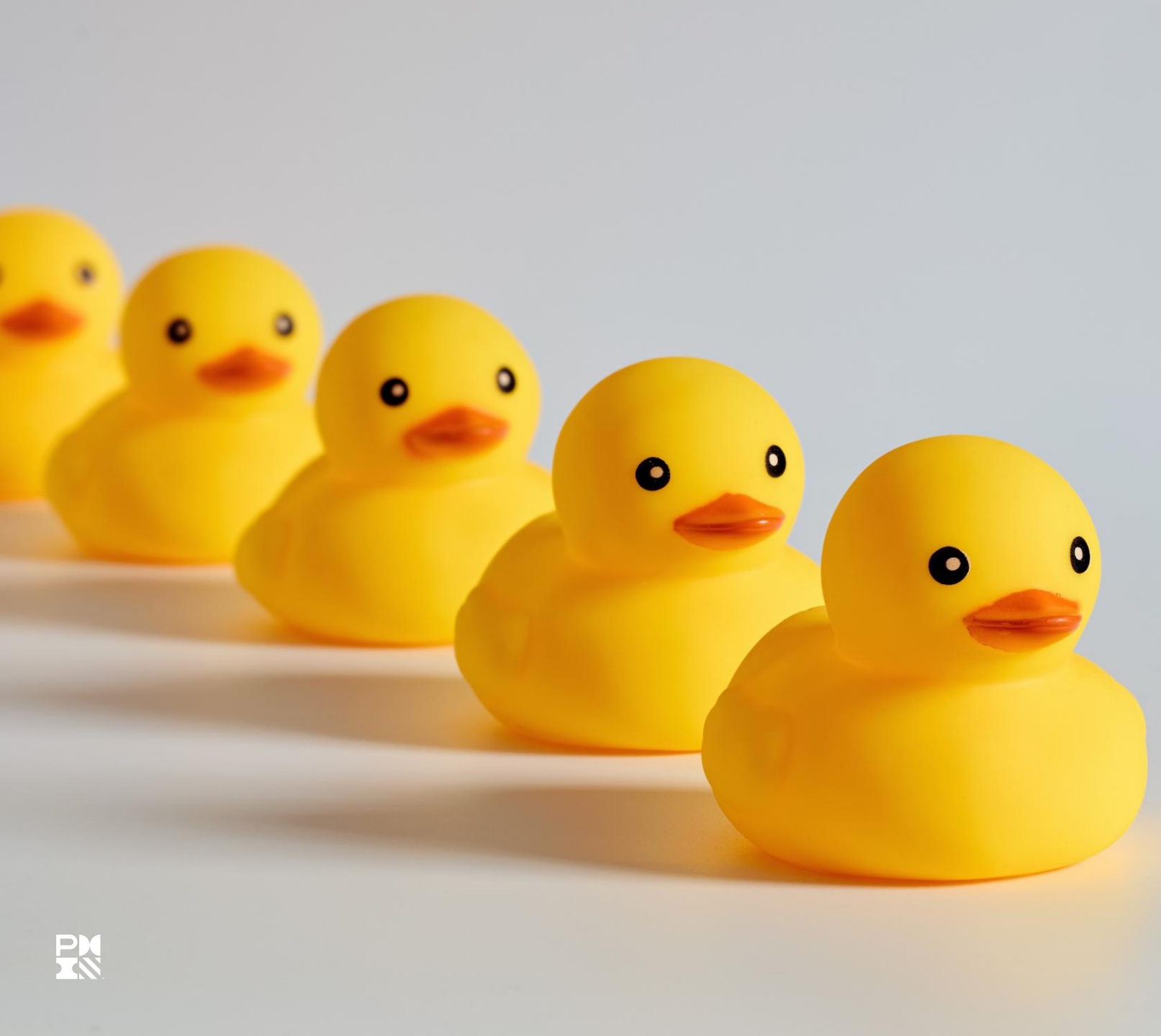
Use lessons learned to avoid risks/impediments recurrence

Section 5 of 5

Lessons learned



Lessons learned is the knowledge gained during a project, which shows how project events were addressed or should be addressed in the future, for the purpose of improving future performance.



Use lessons learned to avoid recurrence

If another team has faced and overcome a similar risk in the past, we may be able to use their strategy to overcome our own risks

Review lessons learned from similar and parallel projects

Past similar or parallel projects may have solved risks your project faces





Ask team for solutions

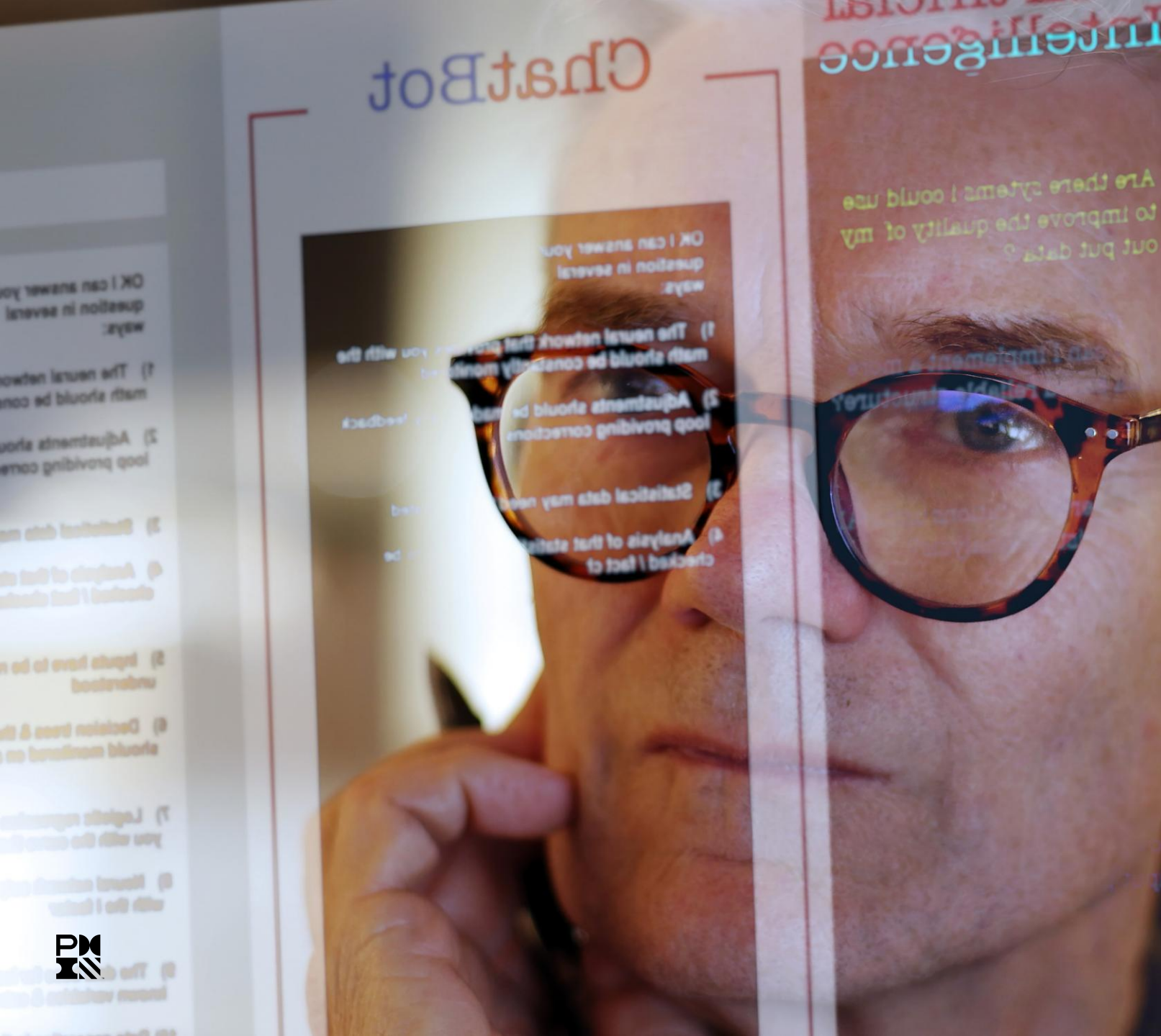
Their responses may
provide a sort of tacit
lessons learned

Use communities of practice

Comes together to learn from each other and develop their skills and knowledge

Can be ideal places to share and understand lessons learned





Artificial intelligence

You can ask an AI chatbot to find potential risks and lessons learned that are pertinent to your domain

Be careful about the information you share with a chatbot

Knowledge check

What are lessons learned?

A. Problems from prior projects

B. Problems faced by another team

C. A way to develop your team's skills and knowledge

D. The knowledge gained during a project

Lessons learned are **the knowledge gained during a project**, which shows how project events were addressed or should be addressed in the future, for the purpose of improving future performance.



**Visualize the
end-to-end
flow of value in
the system**



**Use metrics,
tools, and
feedback loops
to identify
waste**



**Prioritize
waste-
reduction
activities**



**Iterate on
identification
and reduction
of waste**



**Seek early
feedback**



**Manage
agile metrics**



**Manage
impediments
and risk**



**Recognize and
eliminate
waste**



**Perform
continuous
improvements**



**Engage
customers**



**Optimize
flow**



Visualize the end-to-end flow of value in the system

Section 1 of 4

Definitions

Value-added time is the time during the cycle that value is being added.

Non-value-added time is all the rest of the cycle, during which we find delays, waste, and constraints.

Total cycle time is the sum of a process's value-added and nonvalue-added times.

Estimate time for each step



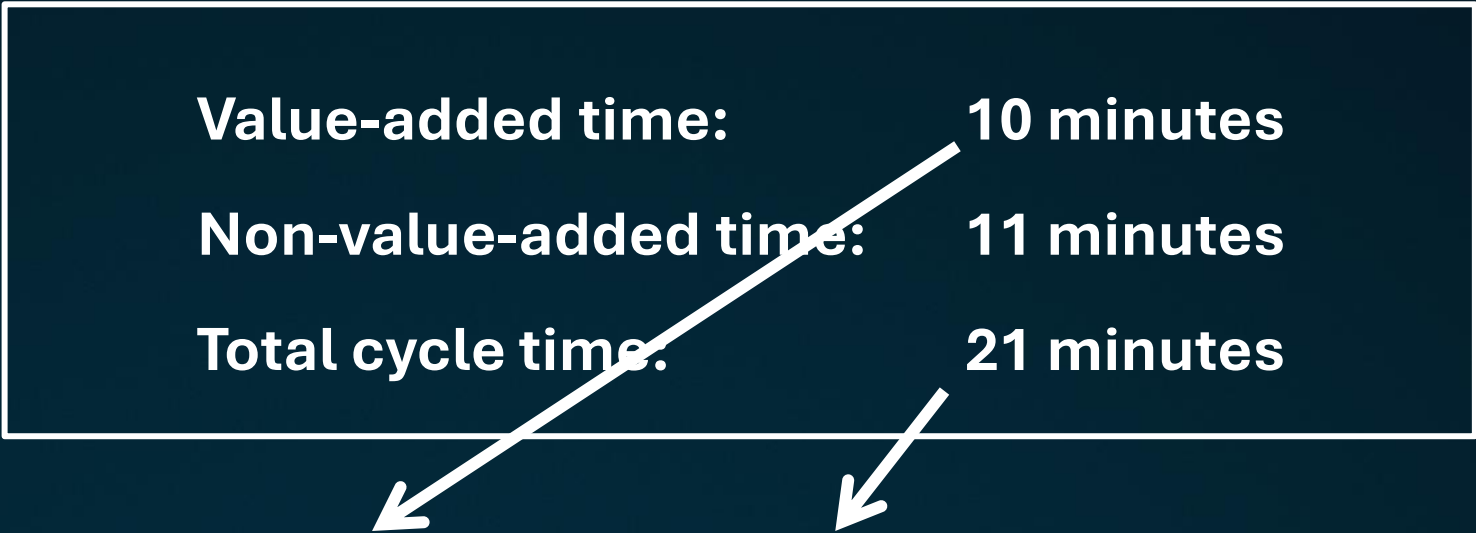
Value-added or non-value-added?



Process efficiency

$$\text{Process efficiency} = \text{Value-added time} \div \text{Total cycle time}$$

Value-added time:	10 minutes
Non-value-added time:	11 minutes
Total cycle time:	21 minutes



$$10 \text{ minutes} \div 21 \text{ minutes} = 48\%$$

Optimizing the process

We need to reduce or,
if possible, eliminate
wait times and loopbacks



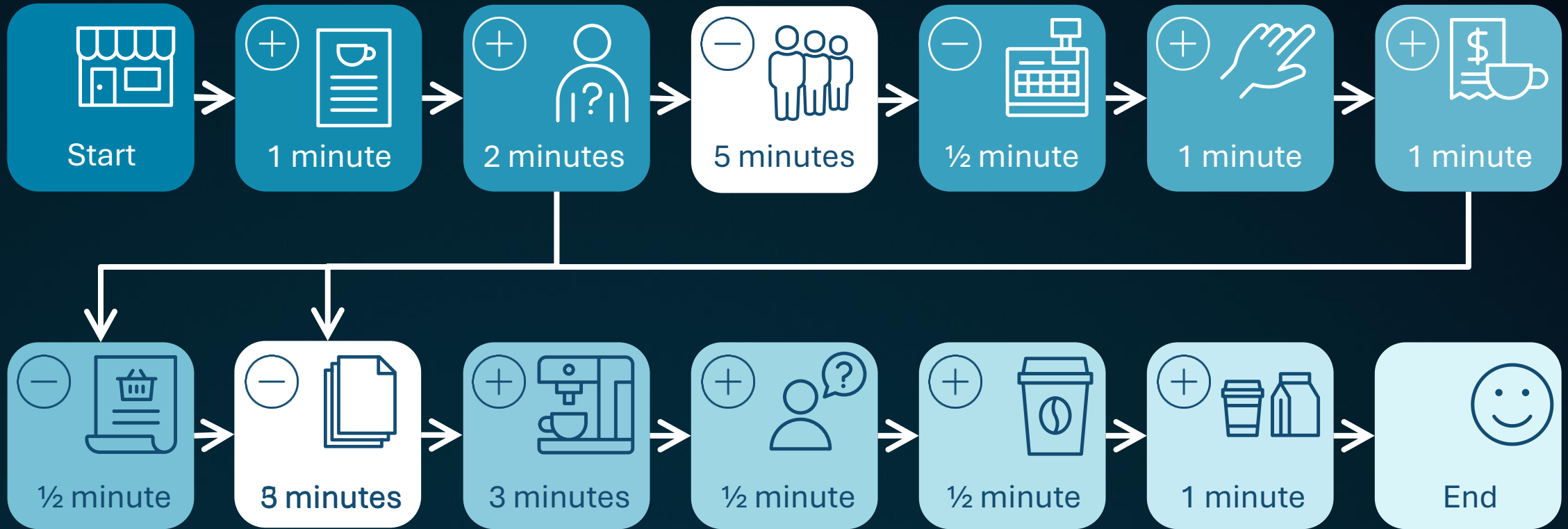


Developing an app

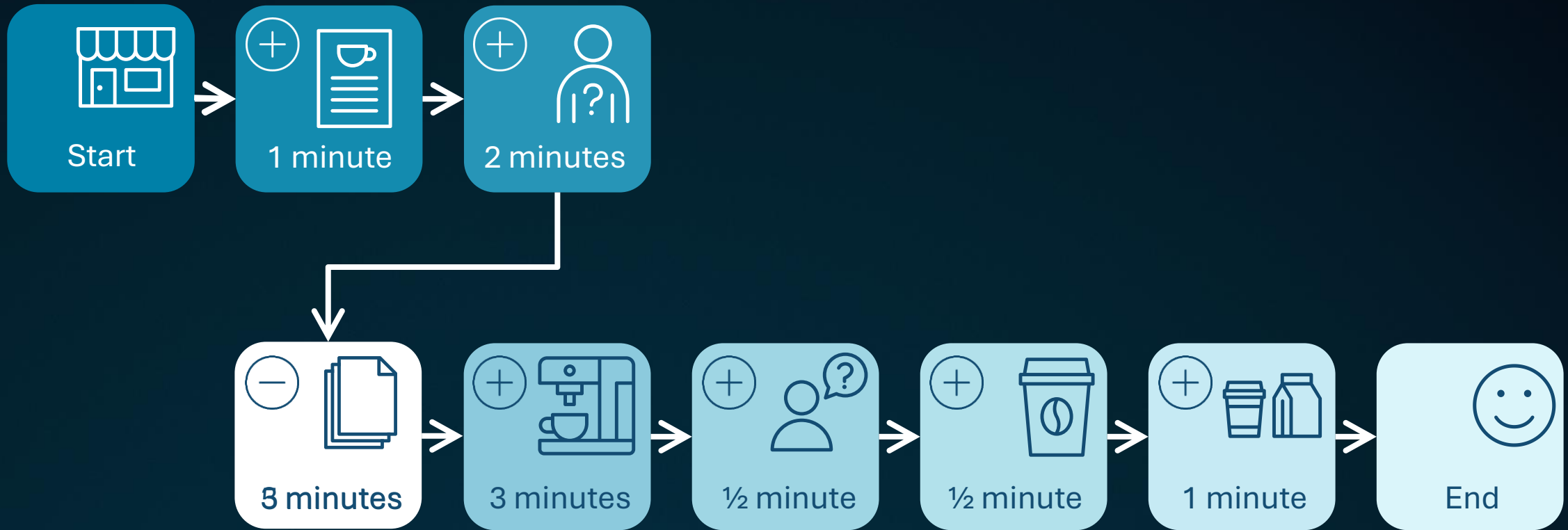
This won't eliminate people wandering into the coffee shop and ordering something in person

It should significantly reduce the wait to order and pay for a drink

Walk-in value stream

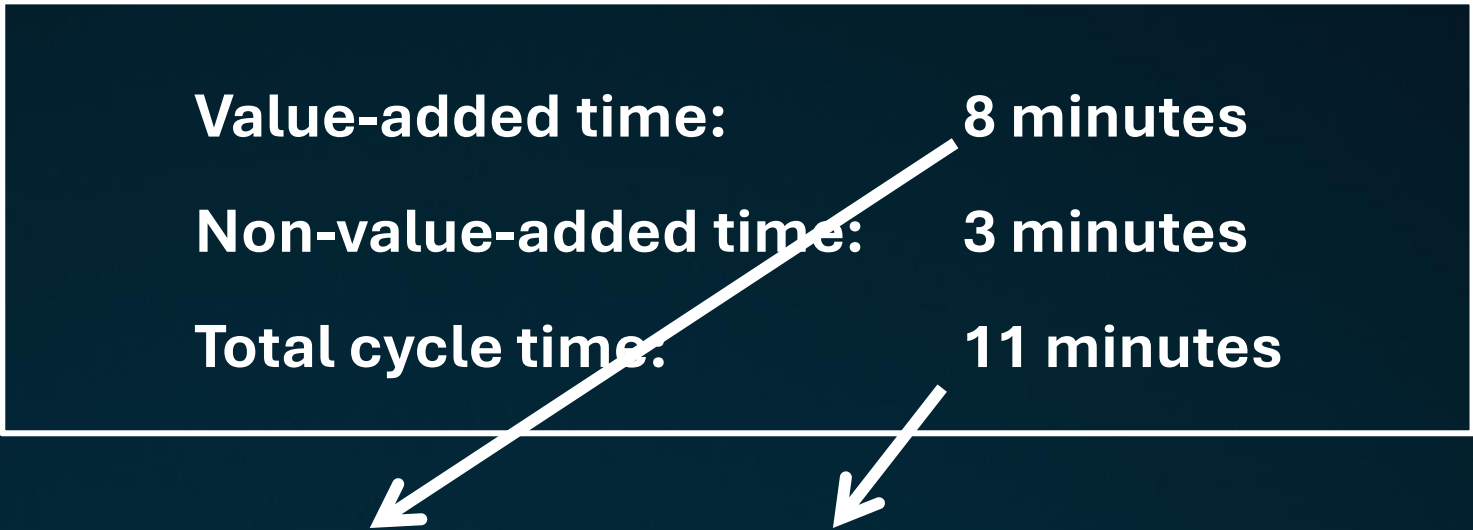


Order Ahead App value stream



App process efficiency

Value-added time:	8 minutes
Non-value-added time:	3 minutes
Total cycle time:	11 minutes


$$8 \text{ minutes} \div 11 \text{ minutes} = 73\%$$



Use metrics, tools, and feedback loops to identify waste

Section 2 of 4

Metrics considerations



Fit for purpose



Specific



Aligned



Consistent



Actionable



Timely

Useful metrics categories



Queue length



Work in progress



Throughput



Process efficiency



Queue length

Can provide insight into the amount of work in progress and the rate of work completion

- Cycle time
- Lead time
- Queue time
- Batch size

Work in progress

The amount of work currently being worked on that has not been completed

- Estimate at completion (EAC)
- Ratio of story points completed to the total story points planned





Throughput

The rate at which the team can complete work items over a given period

- Features complete versus features remaining
- Velocity

Process efficiency

The team's ability to deliver value to the customer while minimizing waste and maximizing productivity

- Cycle time
- Lead time
- Velocity



Tools



**Value stream
mapping**

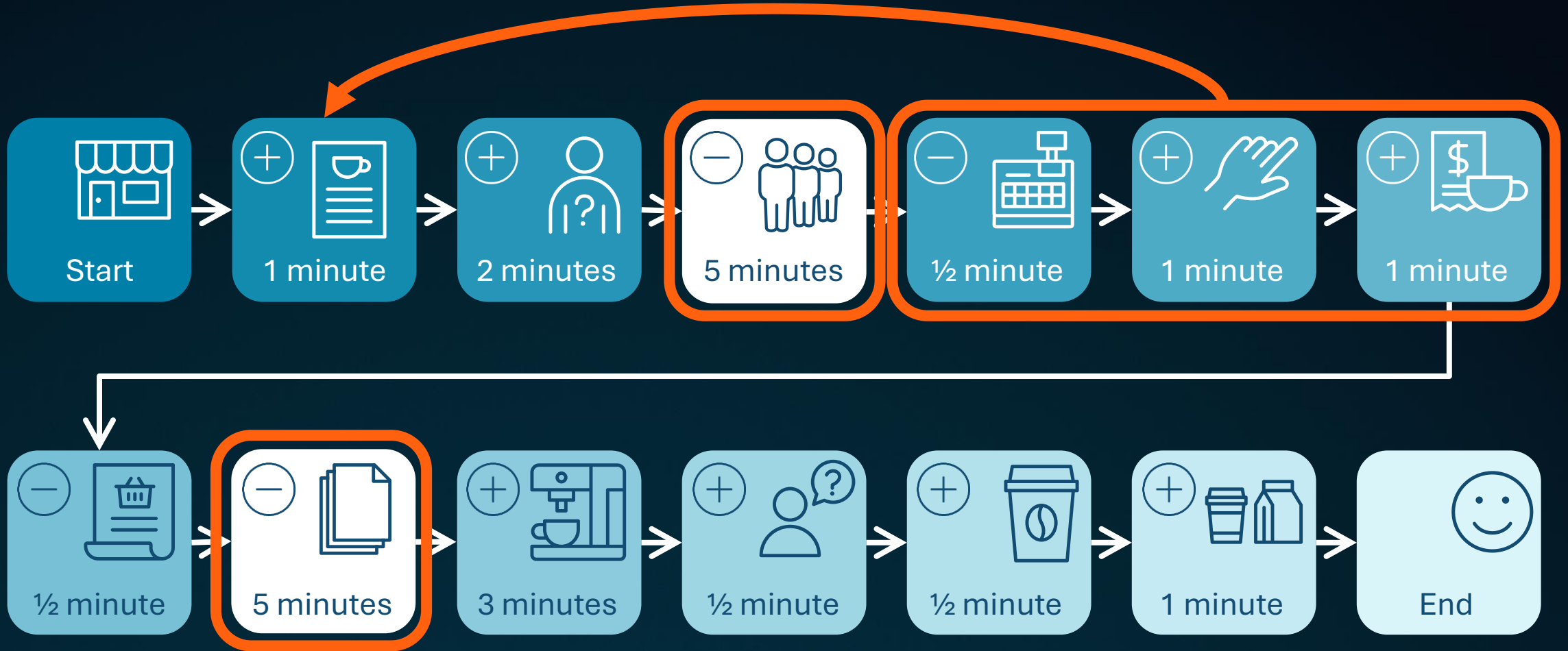


**Theory of
constraints**



**Cumulative flow
diagram analysis**

Value stream map analysis



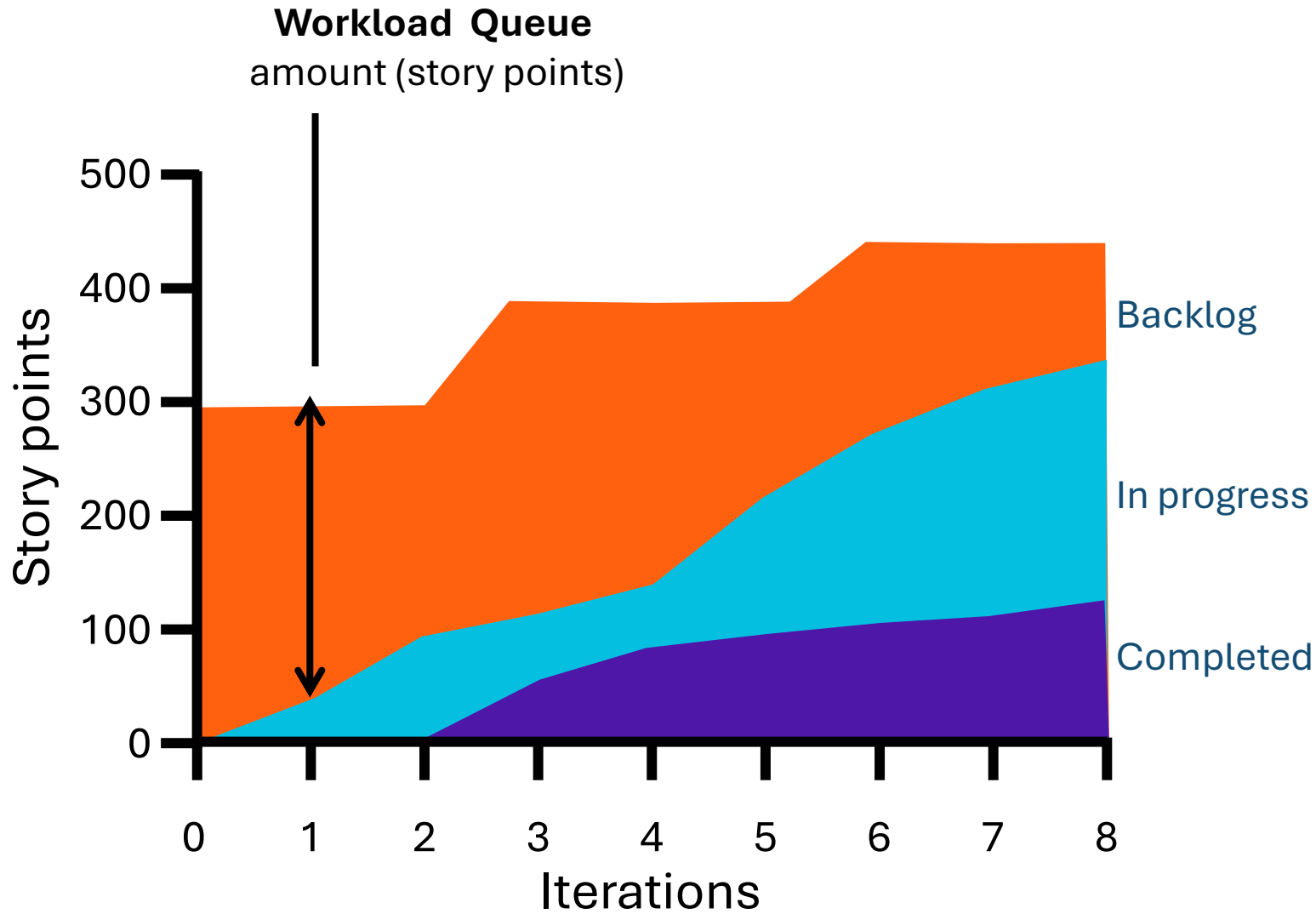
Theory of constraints

Throughput can only be improved when the constraint is improved

Spending time optimizing nonconstraints will not provide significant benefits

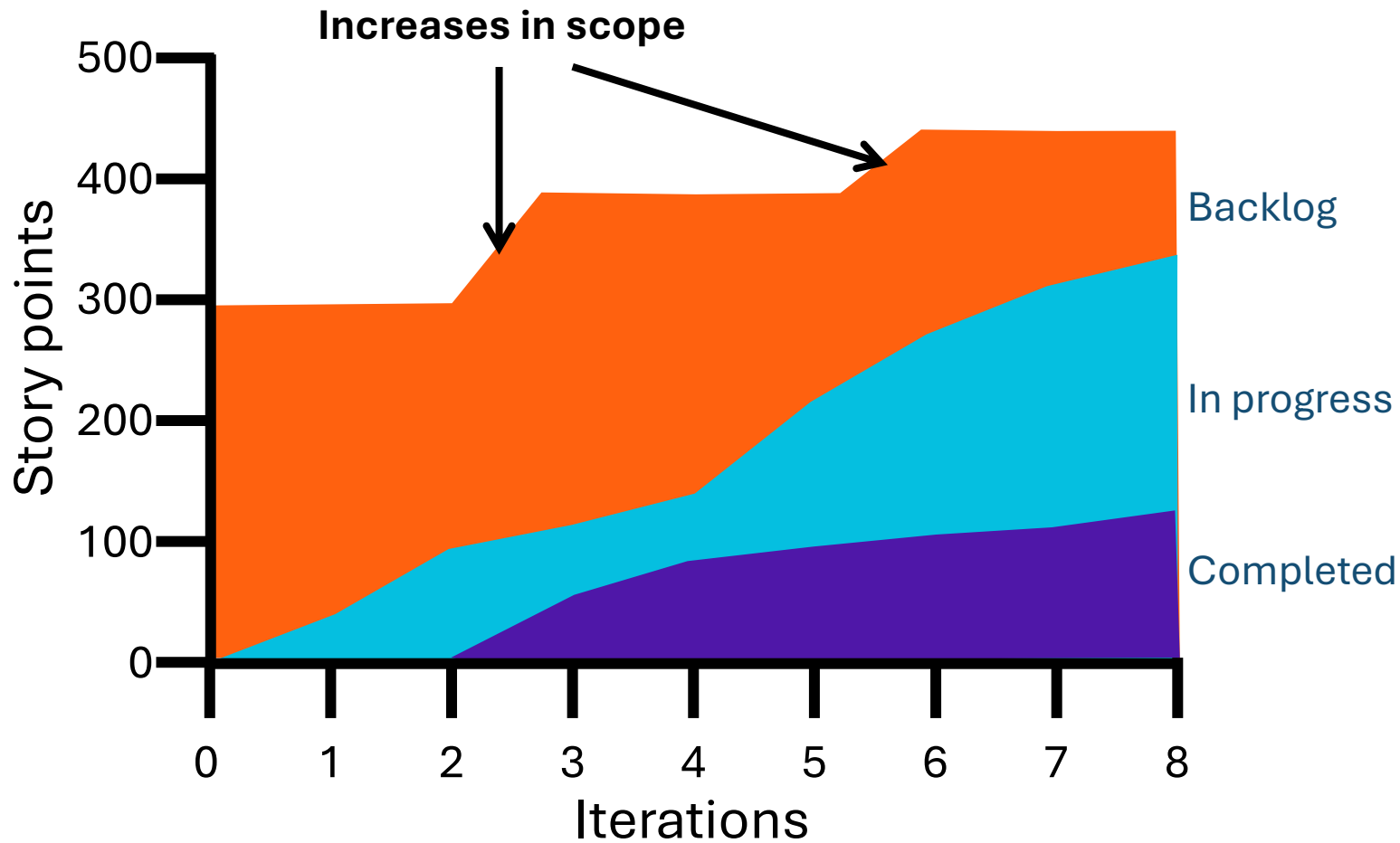


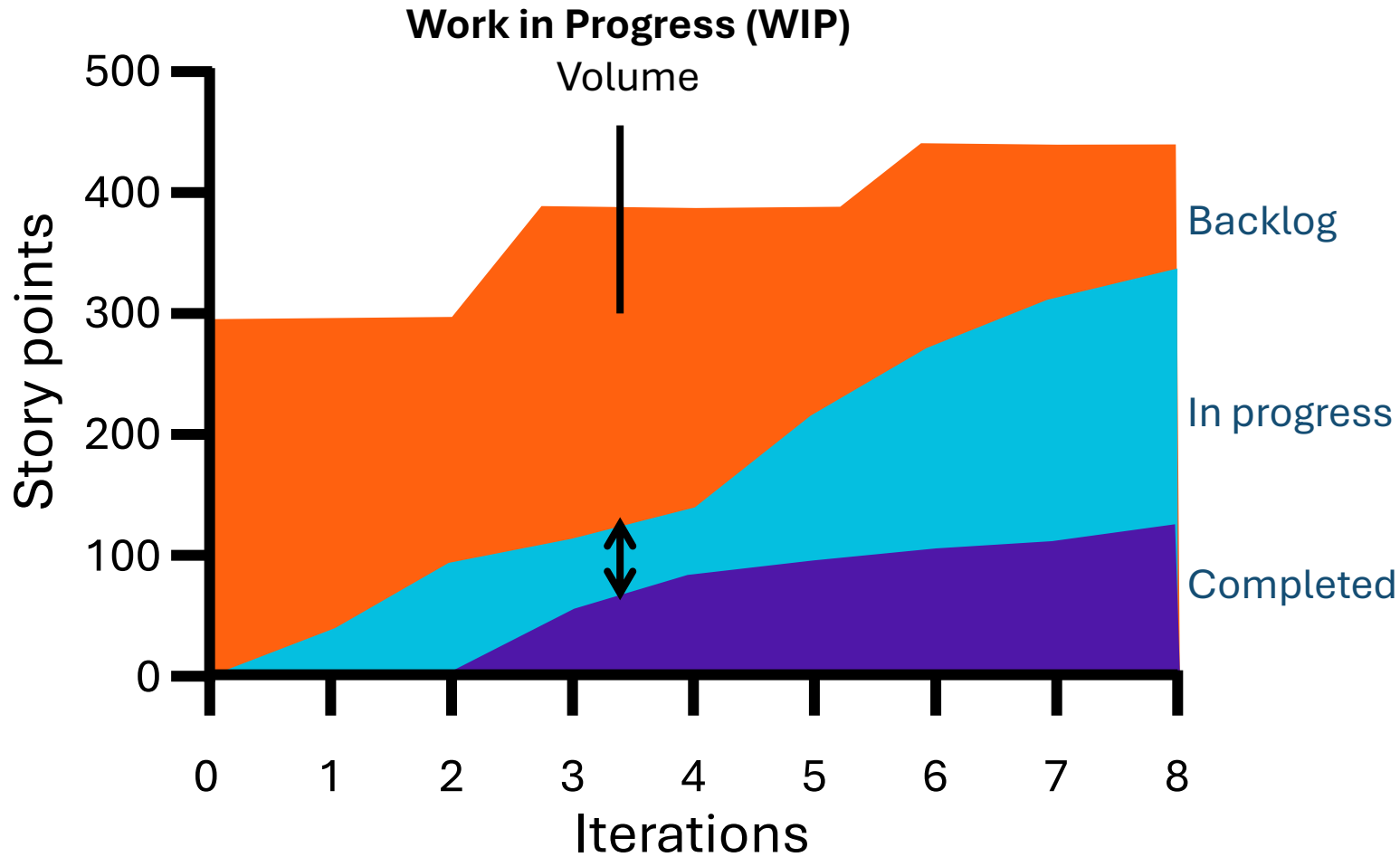
Cumulative flow diagrams: Queues or Backlogs



Cumulative flow diagrams:

Increases in scope





Cumulative flow diagrams:

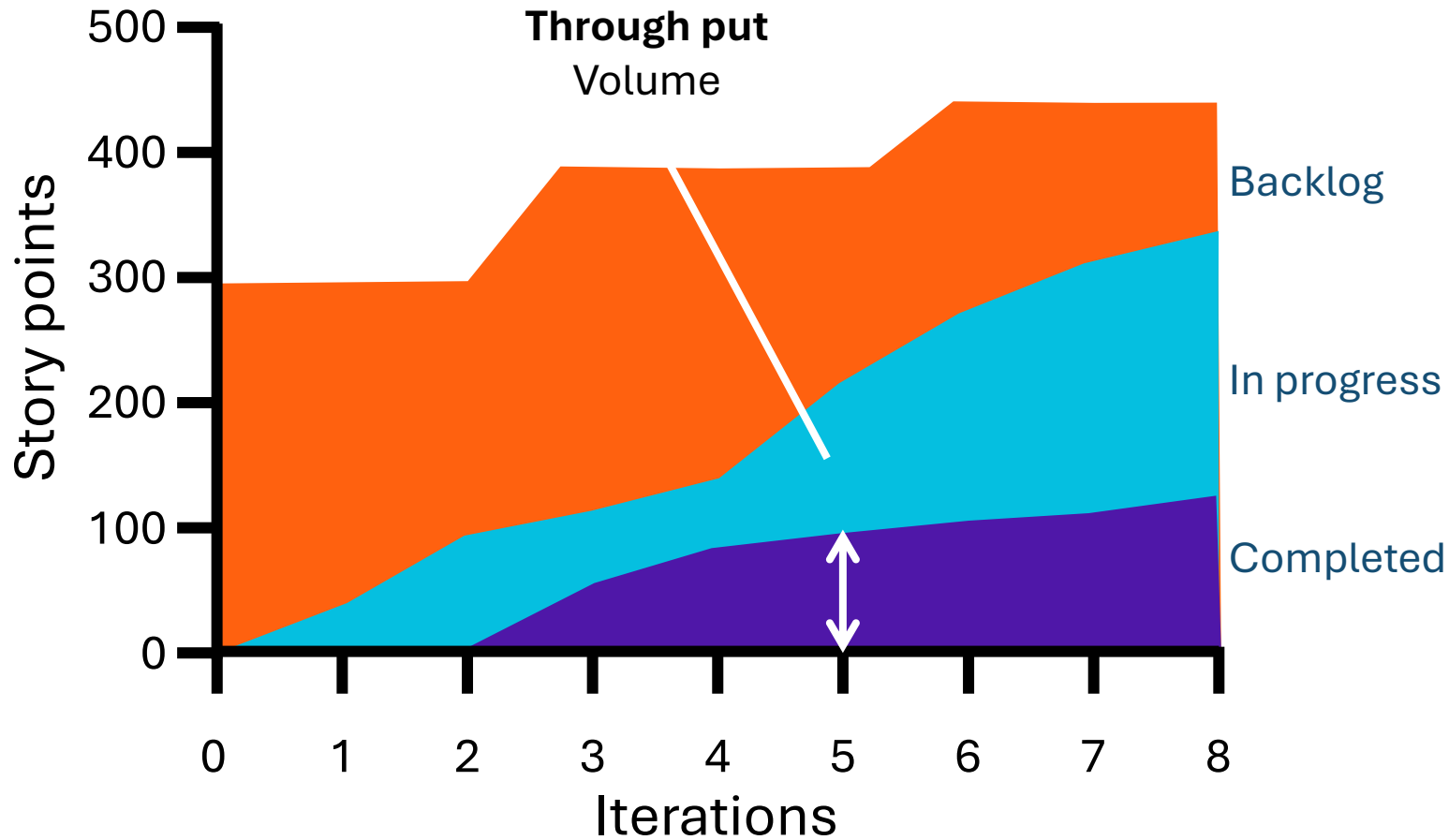
Work in Progress

Limiting work in progress minimizes task switching, improves quality and reduces risk.

Cumulative flow diagrams:

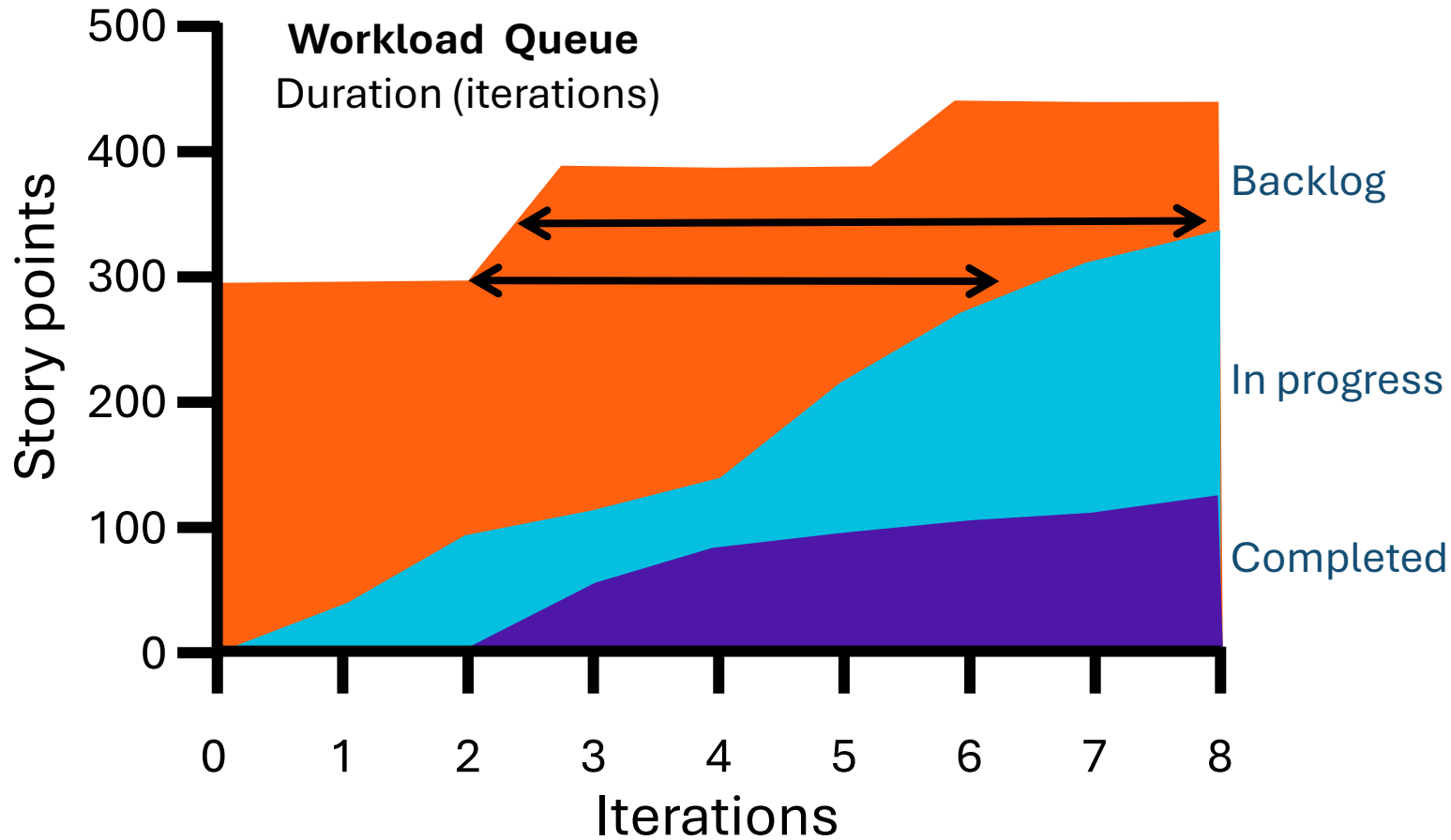
Throughput

Amount of work completed for a set period of time



Cumulative flow diagrams:

Queues or Backlogs

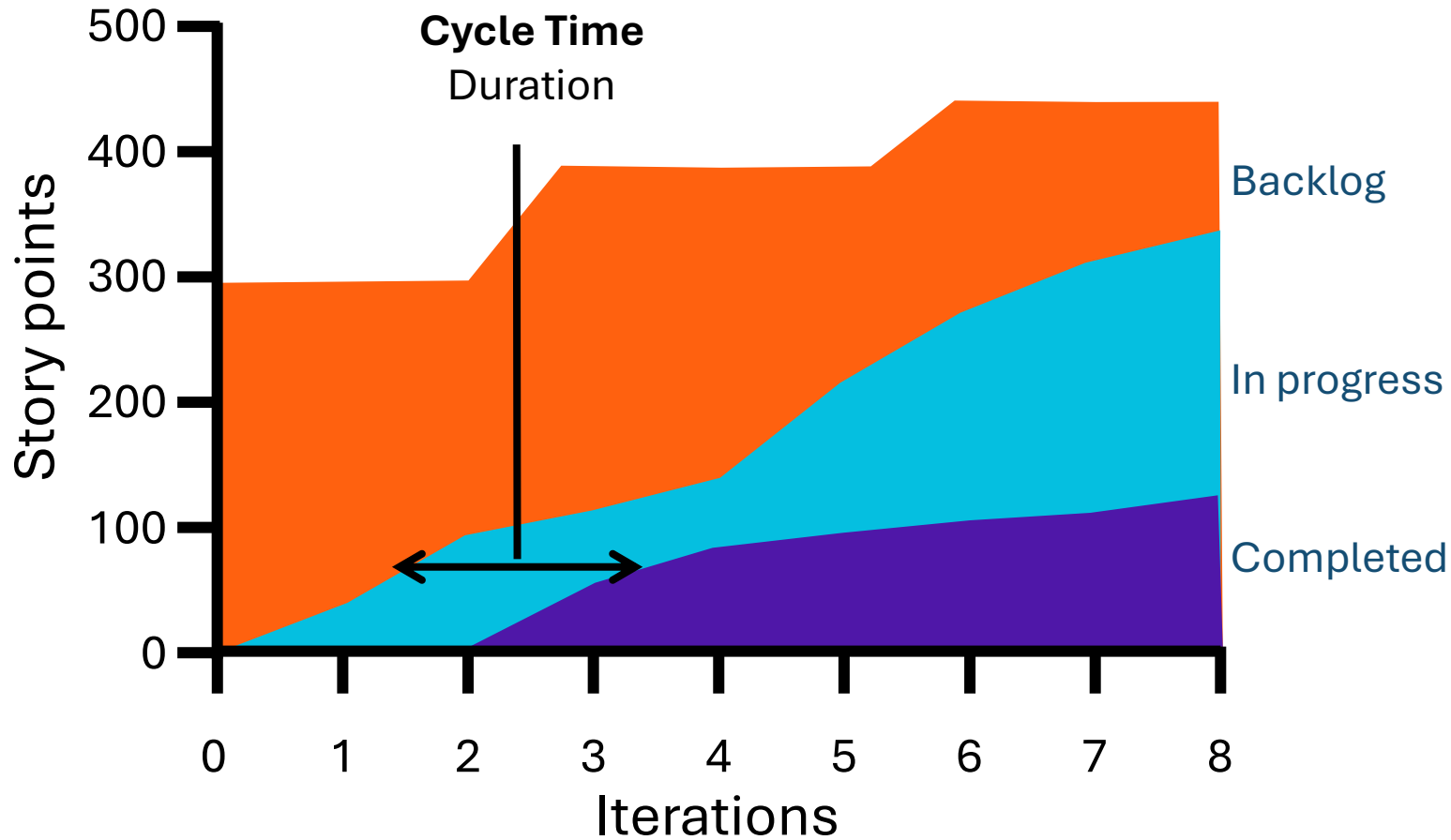


Little's Law: A mathematical formula from queuing theory that proves the duration of a work queue is dependent on its size

Cumulative flow diagrams:

Cycle Time

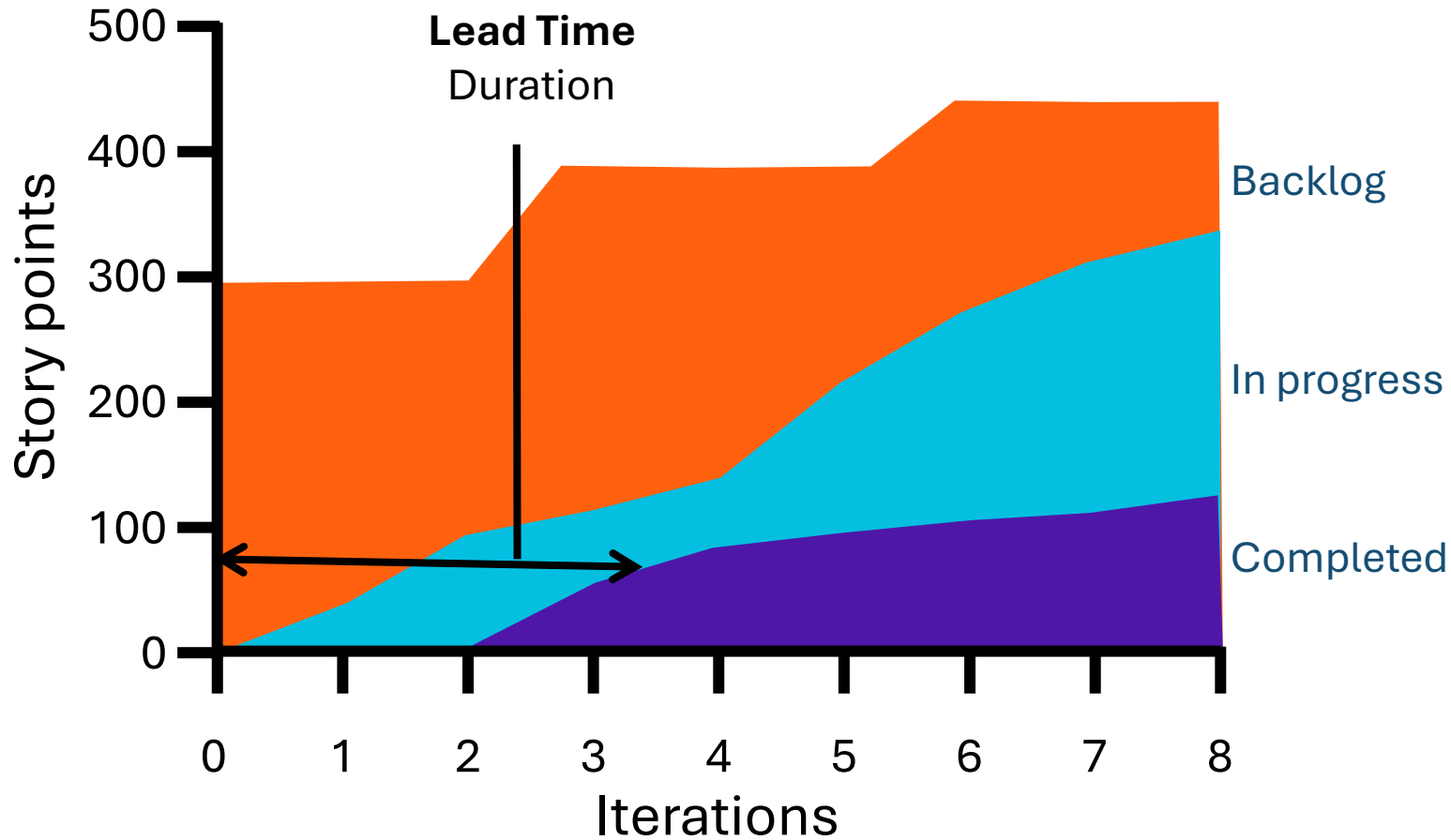
Time from start to completion of a work increment



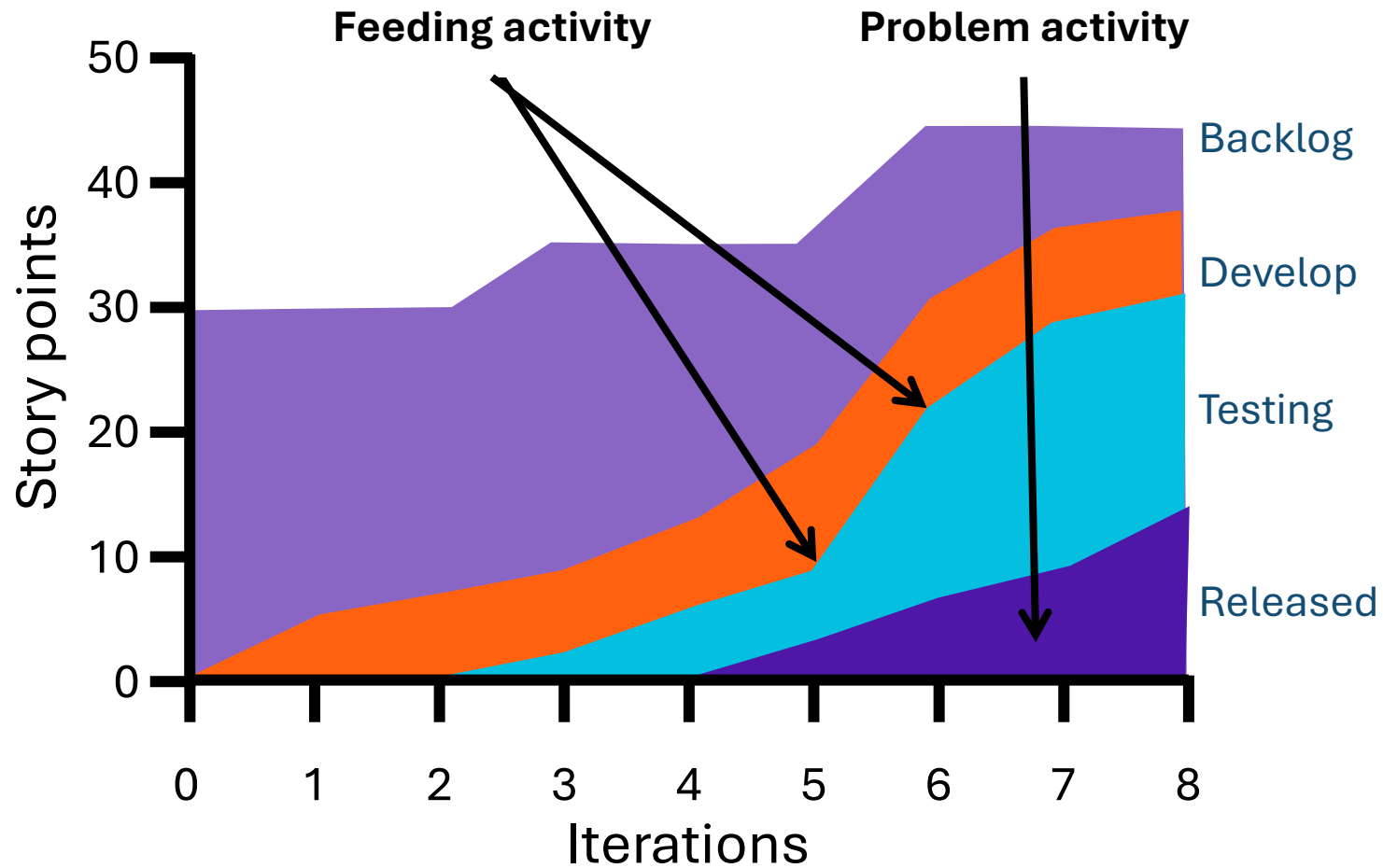
Cumulative flow diagrams:

Cycle Time

Queue time + Cycle time



Using a cumulative flow diagram to identify a bottleneck



Knowledge check

True or false: Queue length can provide insight into the amount of work in progress and the rate of work completion



True



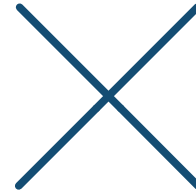
False

Knowledge check

True or false: According to the theory of constraints, spending time optimizing nonconstraints can still provide significant benefits



True



False



Prioritize waste-reduction activities

Section 3 of 4



Team selects and works on waste-reduction activities

Can you identify a waste reduction activity that would help your project?

Putting waste-reduction tasks in the backlog



Feature

Waste-
reduction task

Feature

Feature

Waste-
reduction task

Feature

**How do we put a value on
waste-reduction tasks
so the product owner
can prioritize them?**

Estimating the value of waste-reduction tasks



**Determine the
total value of
the waste**



**Calculate the
potential savings**



**Calculate the
return on
investment**



**Prioritize
the task**

Work with the product owner to set priorities

You may need to explain why waste-reduction activities are important



Review the results

If it was effective, the team moves on to the next task

If it was not, more work may be necessary



Knowledge check

How do you estimate the value of a waste-reduction task?

- A. Calculate the return on investment, determine the total value of the waste, calculate the potential savings, prioritize the task, calculate the return on investment
- B. Determine the total value of the waste, calculate the potential savings, calculate the return on investment, prioritize the task**
- C. Calculate the potential savings, prioritize the task, calculate the return on investment, determine the total value of the waste

To estimate the value of a waste-reduction task, you first determine the total value of the waste, then calculate the potential savings, then calculate the return on investment, and finally prioritize the task based on the estimated return on investment



Iterate on identification and reduction of waste

Section 4 of 4

Theory of constraints

Provides precise and sustained focus on improving the current constraint

Once the constraint is improved, all that focus moves to the next constraint



Sources:
Goldratt, E. M., & Cox, J. (1984). *The goal*. North River Press, Inc.
Goldratt, E. M. (2017). *Critical chain*. Routledge.

© 2024 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI course.

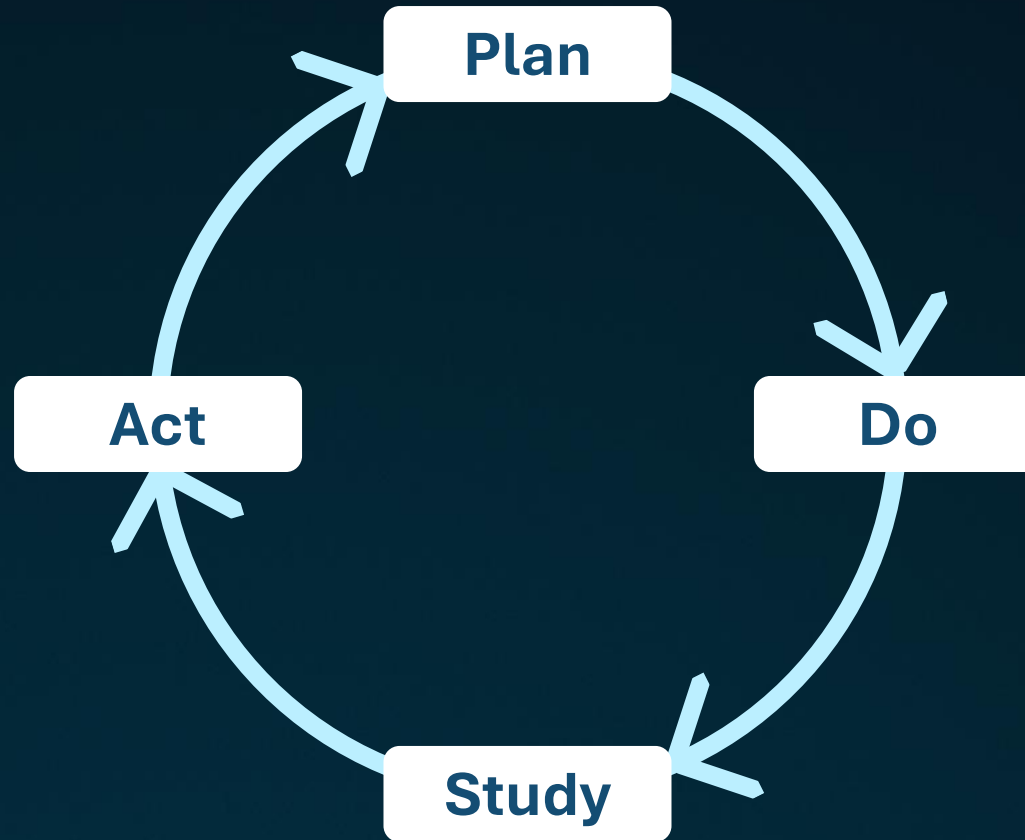


Quality loop

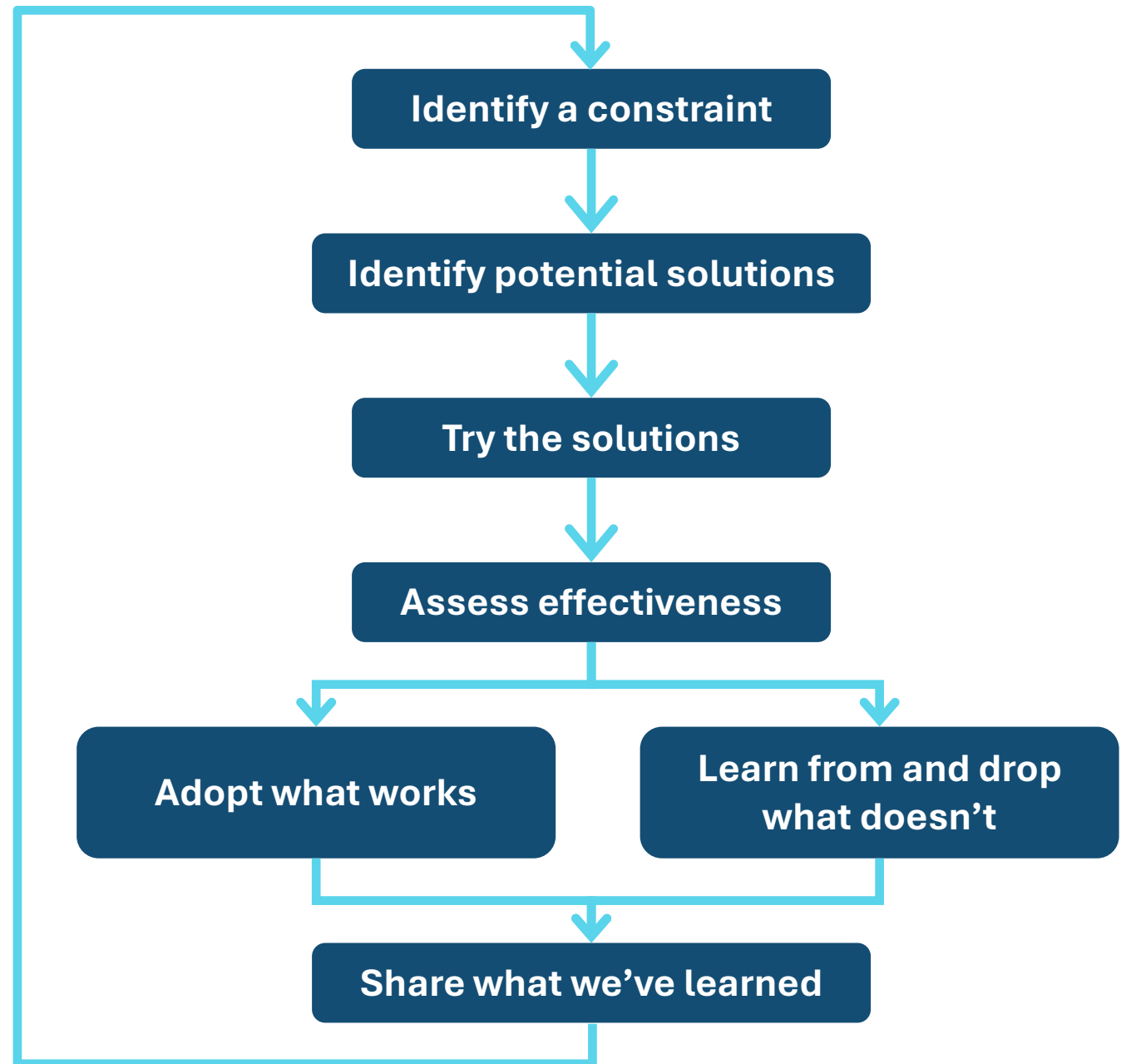
Expose the next weakest link in the chain, strengthen it, and move to the next one

Don't let inertia become a constraint to improvement

Deming / Shwartz Cycle



Improving constraints through experiments

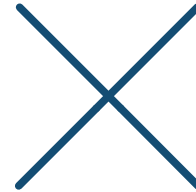


Knowledge check

True or false: In Deming's PDCA Cycle, the letters stand for prepare, do, study, and act



True



False

Knowledge check

True or false: The theory of constraints provides precise and sustained focus on improving the current constraint



True



False

DAILY PMP BOOTCAMP SURVEY



- **Our goal** is to provide the best possible Bootcamp experience for a live streaming webinar, with hundreds of participants.
- For each Bootcamp session,
 - Let us know **what you liked** about the experience – your comments really matter.
 - Please include a thank you **to the mentor(s)** working off camera.
 - If you have **recommendations**, share those too!

We sincerely value your opinion!

LOOK FOR THE SURVEY LINK IN THE CHAT

Survey Scale

- This Scale: 0 not at all likely- 10 extremely likely



On a scale of 0-10, how likely are you to recommend this bootcamp to someone else?

This Scale: 0 not at all likely - 10 extremely likely

0	1	2	3	4	5	6	7	8	9	10
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐